# HiveGuard

## A system for early detection of mold in beehives using IoT

Mbada Shehady and Tamer Amer
Advisor: Dr. Naomi Unkelos-Shpigel

## 1 Introduction

- Mold is a critical problem in beekeeping, especially during winter, reducing hive productivity and threatening colony survival.
- High humidity and poor airflow inside hives create conditions for mold growth, contaminating honey and pollen.
- This project introduces an IoT-based system that combines sensors, real-time monitoring, and automated ventilation to prevent mold.
- By analyzing temperature and humidity data, the system delivers alerts and insights to support beekeepers in protecting their colonies.
- A smart IoT-based solution reduces the need for chemical antifungals, supporting eco-friendly beekeeping.





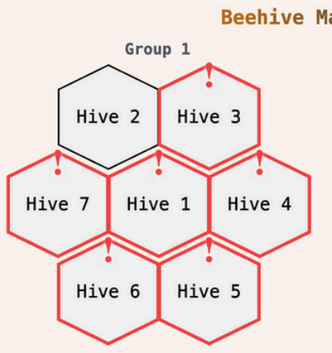
## 2 Impact on Beekeeping Practices

1.Reduces colony stress by minimizing mold-related cleaning behaviors.
2. Improves winter survival rates through better microclimate control.
3. Saves resources by preventing loss of honey and pollen to mold.
4. Supports sustainable beekeeping with solar-powered monitoring.
5. Enables remote hive management, reducing manual inspections.



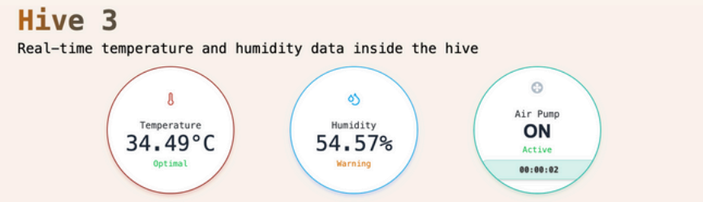
## 3 Our Solution

- Sensors track humidity & temperature inside the hive.
- 4G Portable Modem for wifi connection & MQTT transmit data to the cloud.
- Automated air pump activates when unsafe conditions appear.
- Dashboard shows real-time graphs, historical trends, and hive comparisons.
- Solar-powered & sustainable, designed for remote beekeeping.
















## 4 Evaluation

**Beekeepers tested HiveGuard and provided positive feedback:**

- “Yes, because it reduces manual effort”
- “Yes always good to know whats happening inside the beehives”
- “Yes, because it saves time and lets me check hives without visiting them physically”

**Future improvements:**

- Location map
- Camera to watch hive entrance activity
- Battery status

